

Electrodynamics(H2) (SC1.102a)
IIIT-H, Semester Winter 26, Assignment 1

Submission deadline: 6th April 2026

1. Consider a straight line segment of length L carrying a uniform line charge density λ . At one end of the line, and in the direction perpendicular to the line there is a point P at a distance z . Find the electric field (both magnitude and direction) at P . Also calculate the potential and from that obtain the field. *Solve it by using Coulomb's law.*
2. Find the electric field at a distance z above the center of a flat circular disk of radius R and carrying uniform surface charge density σ . Make sure that your formula makes sense for $R \rightarrow \infty$ as well as $z \gg R$. *Solve using Coulomb's law.*
3. Use Gauss's law to find the field inside and outside an infinitely long hollow cylindrical tube, which carries a uniform surface charge σ .
4. Consider a surface with surface charge density σ . Denote the potential above and below the surface as V_a and V_b . Using Gauss's law show that

$$V_a = V_b, \quad \nabla V_a - \nabla V_b = -\frac{1}{\epsilon_0} \sigma \hat{\mathbf{n}}.$$

5. Find the energy stored in a uniformly charged hollow cylinder of radius R and length ℓ .
6. Find the capacitor per unit length of two coaxial metal cylindrical tubes of radii a and b .
7. In class we have shown that away from a finite charge distribution in a given volume the, potential is given in terms of an expansion in multiple terms. The dipole term of the expansion reads

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}} \cdot \int \mathbf{r}' \rho(\mathbf{r}') d^3\mathbf{r}'.$$

Starting from this formula show that for a single dipole of dipole moment $\mathbf{p}$ the formula simplifies to

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2}.$$

8. A sphere of radius carries a polarization $\mathbf{P} = k\mathbf{r}$ where k is a constant and $\mathbf{r}$ is a vector from the center. Calculate a) the bound charges σ_b and ρ_b , and then find the electric field outside the sphere.
9. Show that if a dipole is placed in an uniform electric field $\mathbf{E}$, it experiences a torque $\mathbf{p} \times \mathbf{E}$ where $\mathbf{p}$ is the dipole moment.